



Shahid Beheshti University

Department of Computer Science
Advanced Data Mining / Deep Learning

Mini Project 1

RNN-1: Multivariate Weather Forecasting

Recurrent Neural Networks on the Jena Climate Dataset

Project Information

Architecture family:	Recurrent Neural Networks
Dataset:	Jena Climate
Project value:	25 points
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Head TA:	Mr. Mobin Nesari
Student:	Alireza Jahanbakhsh
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Notebook Link:	Kaggle Notebook
Project Link:	GitHub Repository

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Data Mining Course

1 Problem and Data

This project studied multi-step temperature forecasting on the Jena Climate dataset from the Max Planck Institute for Biogeochemistry. The raw CSV loaded by the executed notebook contained 420,551 records and 15 columns. Measurements were originally recorded every 10 minutes; the notebook converted them to hourly observations by taking every sixth record, starting at index 5, which produced 70,091 hourly rows. Negative sentinel values below -9000 in the wind-speed and maximum-wind-speed variables were replaced by 0.0.

The forecasting input used all 14 meteorological variables after removing the timestamp: pressure, temperature, potential temperature, dew-point temperature, relative humidity, vapor-pressure variables, specific humidity, water-vapor concentration, air density, wind velocity, maximum wind velocity, and wind direction. The target was T (degC). Each sample used a 72-hour multivariate history and predicted the next 24 hourly temperatures. The split was chronological: 49,063 rows for training (2009-01-01 01:00 to 2014-08-05 01:10), 10,513 for validation, and 10,515 for test. Sliding windows yielded 48,968 training, 10,418 validation, and 10,420 test examples. Feature means and standard deviations were estimated only on the training partition, then applied to validation and test, so normalization did not leak future information.

2 Method

Two non-neural baselines were implemented. Persistence repeated the final observed temperature in the input window across all 24 forecast horizons. Seasonal naive copied the final 24 hourly temperatures from the input, thereby repeating the preceding daily trajectory. The recurrent models were a single-layer LSTM with a dense 24-step output, a single-layer GRU with a dense 24-step output, and an encoder-decoder Seq2Seq model with an LSTM encoder, `RepeatVector(24)`, an LSTM decoder, and a time-distributed dense output. The recurrent hidden size was 64. Models were trained with Adam at learning rate 10^{-3} , MSE loss, batch size 128, up to 20 epochs, and early stopping with patience 4 on validation loss.

LSTM and GRU cells use learned gates to control how information is retained and updated, reducing the short-memory limitation of a plain recurrent cell. The Seq2Seq variant compresses the 72-hour history into an encoder state and expands that state through a recurrent decoder; unlike an autoregressive decoder, the implemented model receives a repeated encoder state rather than previous predicted temperatures. All three models produced the complete 24-hour trajectory in one forward pass.

Evaluation was performed in Celsius after inverse scaling. For each horizon h , MAE and RMSE were computed across test windows, and the mean across all 24 horizons summarized overall forecasting quality. Per-horizon curves were retained because forecast difficulty changes strongly from the first hour to the twenty-fourth hour.

3 Original Contribution

The original contribution was a hypothesis-driven change-aware training experiment. The notebook first diagnosed a failure mode: rapid temperature transitions were associated with higher GRU error. On 109 evaluation windows spaced 96 hours apart, Spearman correlation between transition intensity and error was positive ($\rho = 0.2129$, $p = 0.0262$), supporting rapid-change windows as a meaningful stress regime rather than an arbitrary slice.

The stated hypothesis was: compared with standard GRU training, change-aware weighting will significantly reduce MAE on rapid-change test windows while keeping stable-regime MAE degradation within 5%. For each window, the change score was the mean absolute hourly temperature difference along the path formed by the last observed temperature and the 24 targets. The rapid threshold was the 75th percentile of training scores, 0.9692 °C mean absolute hourly change; it was not estimated from validation or test data. This rule classified 18.5% of full test windows as rapid, with all others forming the stable-regime guardrail.

The intervention up-weighted rapid training windows and normalized weights to mean one. Uniform, mild, and strong settings used rapid factors 1.0, 1.5, and 2.0, respectively. Selection was based only on validation rapid-regime MAE: strong achieved 2.0984 °C, mild 2.1195 °C, and uniform 2.1865 °C, so strong was fixed before the final test comparison. This isolates the weighting rule: model architecture, loss, optimizer, training windows, and training budget otherwise remained unchanged.

Table 1: Change-aware test ablation. The strong setting was selected by validation.

Scheme	Overall MAE	Rapid MAE	Stable MAE	Overall RMSE	h24 MAE
Uniform	1.7832	1.8379	1.7707	2.2772	2.2876
Mild	1.7816	1.8006	1.7773	2.2861	2.2569
Strong	1.7957	1.7949	1.7959	2.3060	2.2670

The selected strong model improved non-overlapping rapid-regime MAE from 2.0824 to 2.0600 °C, a 1.0779% numerical improvement. However, the one-sided paired Wilcoxon test did not support the hypothesis ($p = 0.3631$), and the paired bootstrap 95% confidence interval for selected-minus-uniform MAE was $[-0.1909, 0.1427]$ °C. Stable-regime MAE degraded by 1.4191%, within the 5% guardrail, but the primary rapid-regime improvement was statistically inconclusive. The correct conclusion is therefore negative: the hypothesis was not supported, despite a small numerical gain in the targeted regime.

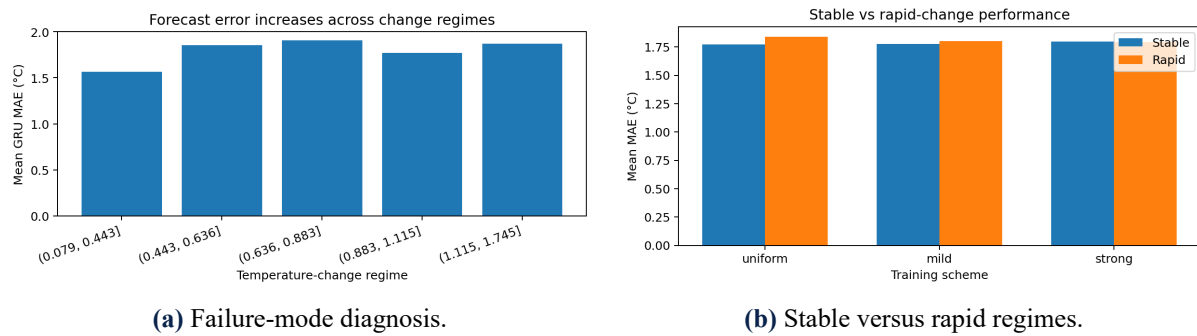


Figure 1: Rapid changes were harder for the GRU, but change-aware weighting produced only small test differences.

4 Experiments

The controlled experiment set included persistence, seasonal naive, LSTM, GRU, and Seq2Seq under the same 72-hour input and 24-hour output setup. The original contribution added the uniform, mild, and strong change-aware GRU ablation. Supplementary experiments addressed the handbook discussion questions: a GRU with additive temporal attention, a compact Transformer encoder compared against the GRU, validation-only data-size and sequence-length regime sweeps, and a three-seed robustness check for the change-aware intervention. The handbook stretch goal was also implemented through residual-based 90% prediction intervals and is reported separately in Section 7.

Table 2: Main test results across 24 forecast horizons.

Model	Mean MAE	Mean RMSE	MAE h=1	MAE h=24
GRU	1.7832	2.2772	0.5542	2.2876
LSTM	1.8211	2.3198	0.6360	2.2798
Seq2Seq	1.8651	2.3669	0.7809	2.3548
Seasonal naive	2.5073	3.2575	2.5066	2.5073
Persistence	3.1574	4.0977	0.6772	2.5073

5 Results and Analysis

The GRU had the best mean MAE and RMSE among the principal models, although LSTM was marginally best at horizon 24. Figure 2 shows why a single average is insufficient. Persistence was competitive only at the first hour, then error compounded rapidly. Seasonal naive was flat by construction and became stronger than persistence at longer horizons. Recurrent models degraded more gradually, indicating that they learned temporal structure beyond copying the last value or the previous day.

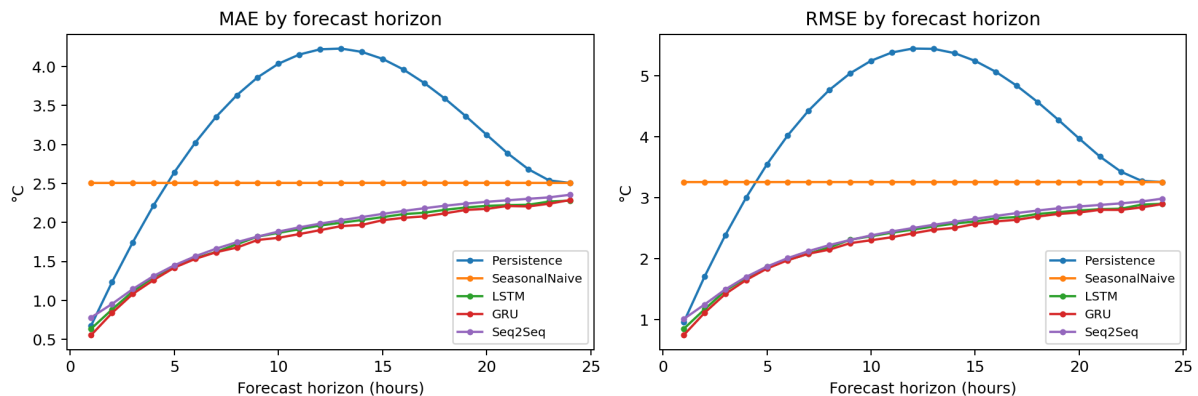


Figure 2: MAE and RMSE by forecast horizon for baselines and recurrent models. Error generally increased with horizon for learned models, while persistence deteriorated sharply after short horizons.

The statistical comparison used sample-wise MAE on the 109 non-overlapping evaluation windows. The Friedman test rejected equality across the five methods ($\chi^2 = 85.8202$, $p = 1.01613 \times 10^{-17}$), after which Holm-adjusted paired Wilcoxon tests localized the differences. The strongest conclusion was not that GRU decisively beat LSTM or Seq2Seq: LSTM–GRU was not significant ($p_{\text{Holm}} = 0.1128$), nor was GRU–Seq2Seq ($p_{\text{Holm}} = 0.0576$). Each recurrent model did significantly outperform each non-neural baseline after adjustment. Persistence versus seasonal naive was itself not significant after adjustment ($p_{\text{Holm}} = 0.0576$). Thus, the reliable result is the recurrent-versus-baseline gap, not the small ranking among recurrent architectures.

Failure analysis showed two consistent patterns. First, forecast error increased with horizon because farther targets are less constrained by the observed state and errors in unresolved weather dynamics accumulate. Second, stronger transitions were associated with larger errors, although the binned full-test means were not strictly monotonic. Figure 3 shows the three highest-MAE forecasts of the selected strong model; these examples illustrate trajectory and amplitude errors that a single aggregate score hides. Multi-seed robustness across seeds 7, 21, and 42 gave a mean rapid-regime change of -0.0499 ± 0.0088 °C (selected minus uniform), but an overall change of $+0.0111 \pm 0.0049$ °C. This repeated the targeted numerical direction while confirming a small average-performance cost; it does not replace the inconclusive pre-specified test.

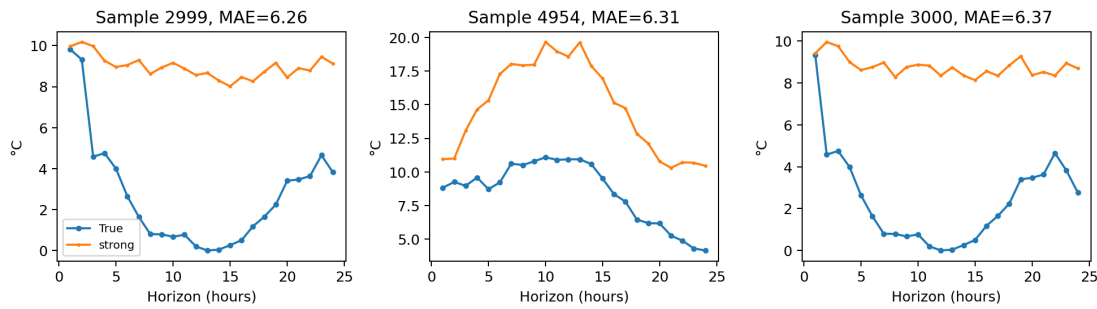


Figure 3: Three highest-MAE forecasts from the validation-selected strong change-aware model. The examples expose substantial trajectory mismatch and support the failure analysis without implying that all large errors are rapid-change events.

6 Discussion and Observation

6.1 Question 1: Attention over Past Timesteps

If temporal attention helps a recurrent forecaster, it should learn to revisit historical timesteps that summarize relevant diurnal context, recent trend changes, and comparable pre-transition patterns. The implemented additive-attention GRU provides direct evidence. Its largest average attention weights occurred around lags 54, 55, 56, 57, and 53 hours before the forecast origin, so the trained model often emphasized observations roughly two days before prediction rather than only the most recent hours.

However, attention did not improve forecasting accuracy in this run. Mean MAE for h1-h6 was 1.1494 °C for GRU+attention versus 1.1145 °C for the plain GRU. Mean MAE for h19-h24 was 2.2507 °C versus 2.2132 °C. The long-horizon one-sided Wilcoxon test gave $p = 0.7063$, so the gain was not larger at long horizons; there was no gain. Attention weights were interpretable as a learned lookup pattern, but visual prominence did not translate into better error.

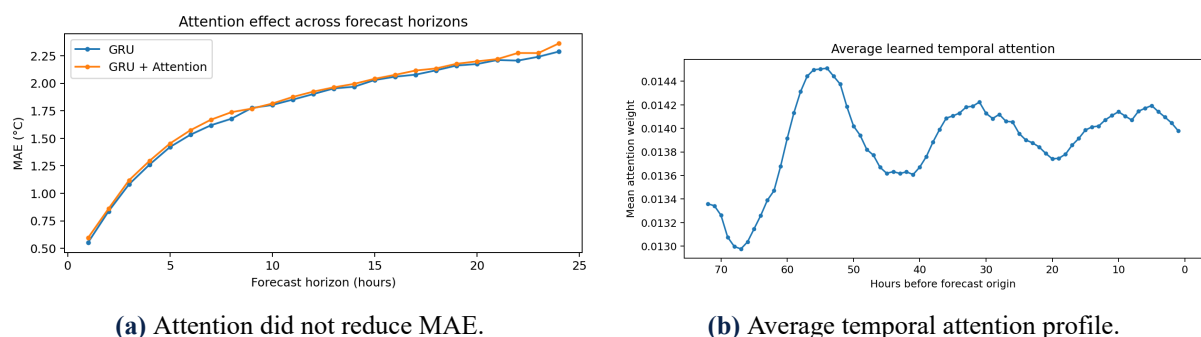


Figure 4: The attention experiment answered the first discussion question empirically: the model learned a nonuniform look-back pattern, but the added mechanism did not improve short- or long-horizon accuracy.

6.2 Question 2: RNN Inductive Bias versus Transformer

The RNN's sequential inductive bias gives the model a compact, ordered state update with shared parameters across time. This encourages local temporal continuity and can be statistically efficient when data and tuning budgets are limited. A Transformer instead compares timesteps directly through self-attention and positional information, offering shorter information paths but less built-in sequential structure. In this notebook, the recurrent bias was useful under the tested budget. The matched comparison gave GRU mean MAE 1.7832 and RMSE 2.2772, while the compact Transformer obtained MAE 2.6733 and RMSE 3.4306. A paired Wilcoxon test on non-overlapping windows reported $p = 9.477 \times 10^{-7}$.

The regime sweep strengthened the conclusion. With 25% of the training data, the GRU had lower validation MAE in all three tested input lengths (24, 72, and 168 hours). With the full training set, it also won all three. The Transformer-minus-GRU validation MAE gap was 0.4643 °C at 24 hours, 1.4916 °C at 72 hours, and 1.8122 °C at 168 hours. Therefore, in this moderate-sized single-station weather setting, the GRU was the safer bet across the tested data-size and context-length regimes. This is not a universal architecture ranking; larger datasets, longer contexts, or a more extensively tuned Transformer could change the result.

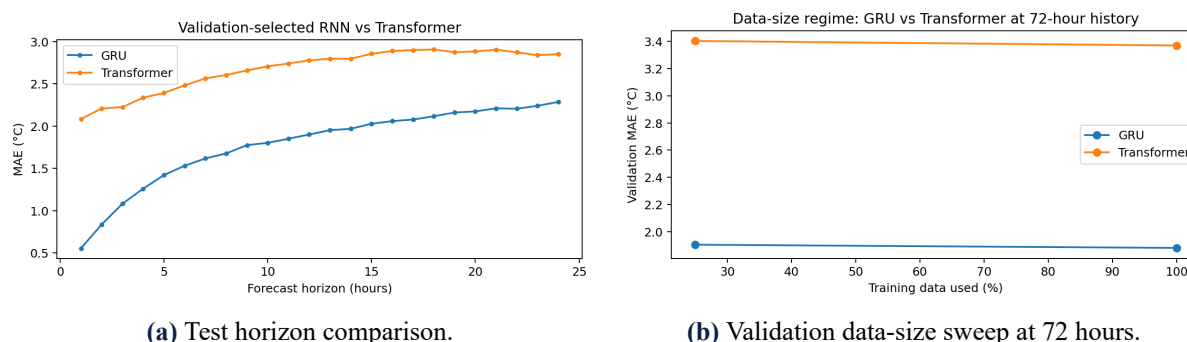


Figure 5: GRU outperformed the compact Transformer under the matched notebook setup and in the tested validation regimes.

7 Bonus / Stretch: Prediction Intervals

The handbook's probabilistic-forecasting stretch goal was implemented without retraining the forecaster. For each horizon, the 5th and 95th percentiles of validation residuals from the validation-selected strong model were added to each test point forecast. This produced horizon-specific empirical 90% intervals while preserving the held-out status of the test targets.

Across all test windows and horizons, empirical coverage was 0.9189 and mean interval width was 8.0395 °C. The horizon-wise interval plot shows wider uncertainty bands at later forecast horizons, consistent with the growth of point-forecast error. These are empirical residual-based intervals calibrated from validation errors, not a learned conditional probabilistic forecasting model.

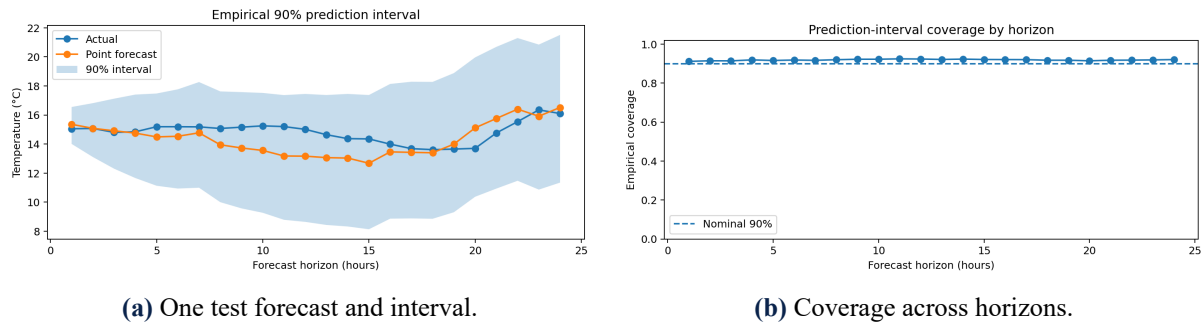


Figure 6: Stretch experiment: horizon-specific 90% intervals calibrated from validation residual quantiles.

8 Limitations and Next Steps

The study used one station, one target variable, hourly subsampling, compact architectures, and a 24-hour forecast horizon. Non-overlapping windows reduced dependence for inference but did not make weather samples fully independent. The Transformer and attention experiments were useful controlled comparisons but were not broad hyperparameter searches. The change-aware intervention was hypothesis-driven and guarded against stable-regime degradation, yet the test evidence was not statistically significant.

Future work should evaluate multi-station generalization, richer change definitions, probabilistic objectives, and longer multi-seed studies. Direct quantile or likelihood training would provide conditional uncertainty rather than relying on pooled validation residuals. The change-aware objective could also use a continuous weighting function and jointly constrain overall MAE, but any such variant should retain validation-only selection and a pre-specified stable-regime guardrail.

9 References

1. Max Planck Institute for Biogeochemistry, *Jena Climate Dataset*. Weather measurements from the Jena, Germany station used in the project.
2. S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
3. K. Cho et al., “Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation,” in *Proceedings of EMNLP*, 2014.